PCBC H Name _____ _____ /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

PRACTICE D · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

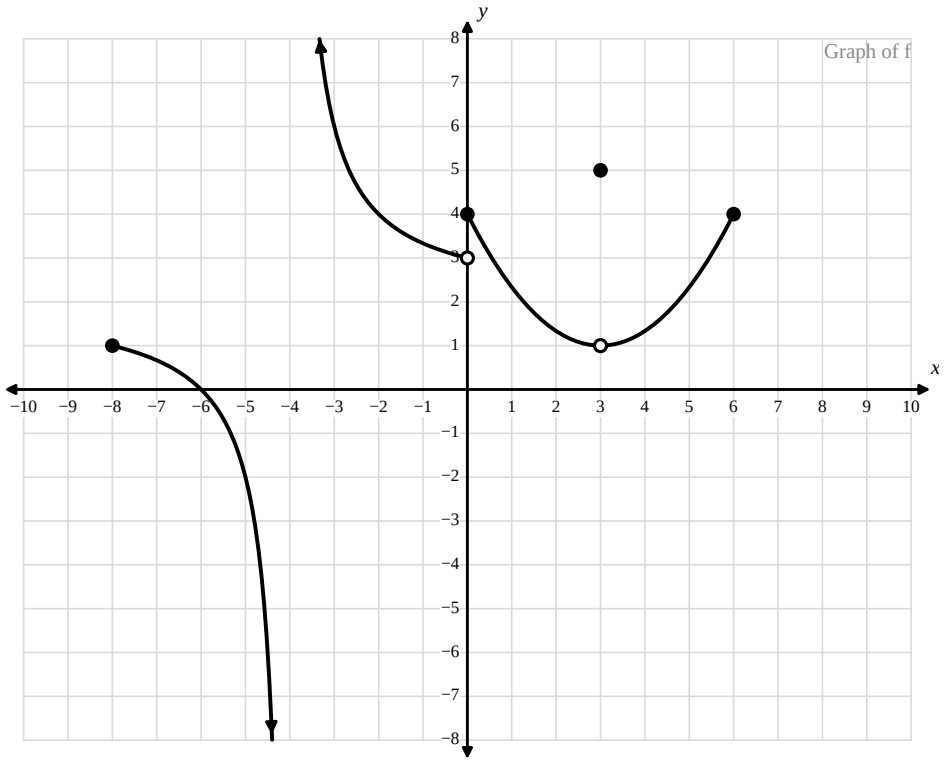
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Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)

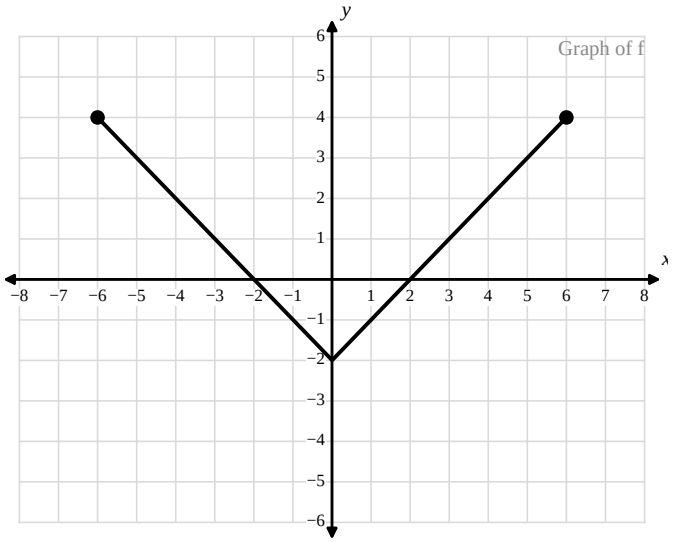


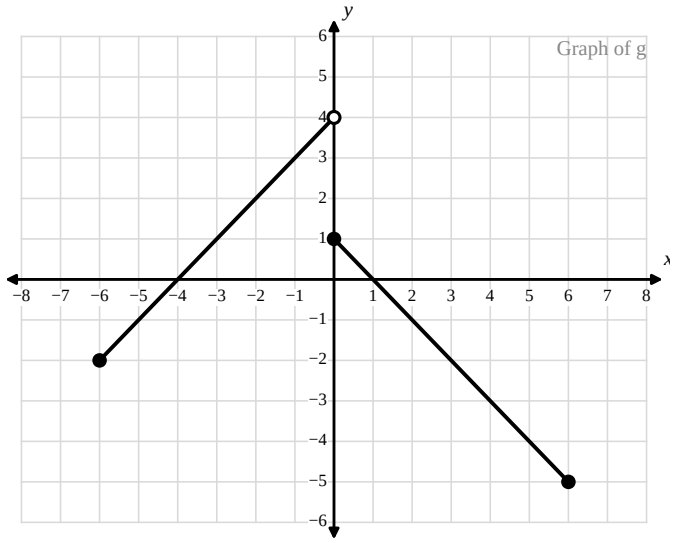
1. $\lim_{x\to 3} f(x) =$	**2.** $\lim_{x\to -4^-} f(x) =$	**3.** $\lim_{x\to 0} f(x) =$

(2 points)

4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = 4$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim\limits_{x \to 0^-} \dfrac{[f(x)]^2 + g(x)}{f(x) - g(x)}$

(2 points)

6. Use the table to estimate $\lim\limits_{x \to 0} k(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	−0.1	−0.01	−0.001	→ 0 ←	0.001	0.01	0.1
$k(x)$	0.951626	0.995017	0.999500		1.000500	1.005017	1.051709

$\lim\limits_{x \to 0^-} k(x) \approx$ _____ $\lim\limits_{x \to 0^+} k(x) \approx$ _____ $\lim\limits_{x \to 0} k(x) \approx$ _____

Evaluate each limit. (2 points each)

7. $\lim\limits_{x\to\frac{\pi}{6}} \left(\sec^2(2x) - 4\sin x\right)$

8. $\lim\limits_{x\to -2} \dfrac{x^3+2x^2-x-2}{x^2+3x+2}$

9. $\lim\limits_{h\to 0} \dfrac{\sqrt{9+h}-3}{h^2+h}$

10. $\lim\limits_{x\to 0} \dfrac{\frac{1}{(x+2)^2}-\frac{1}{4}}{x}$

11. For $f(x)=\dfrac{1}{x^2}$, find $\lim\limits_{h\to 0}\dfrac{f(x+h)-f(x)}{h}$

PCBC H Name **ANSWER KEY** **22** /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

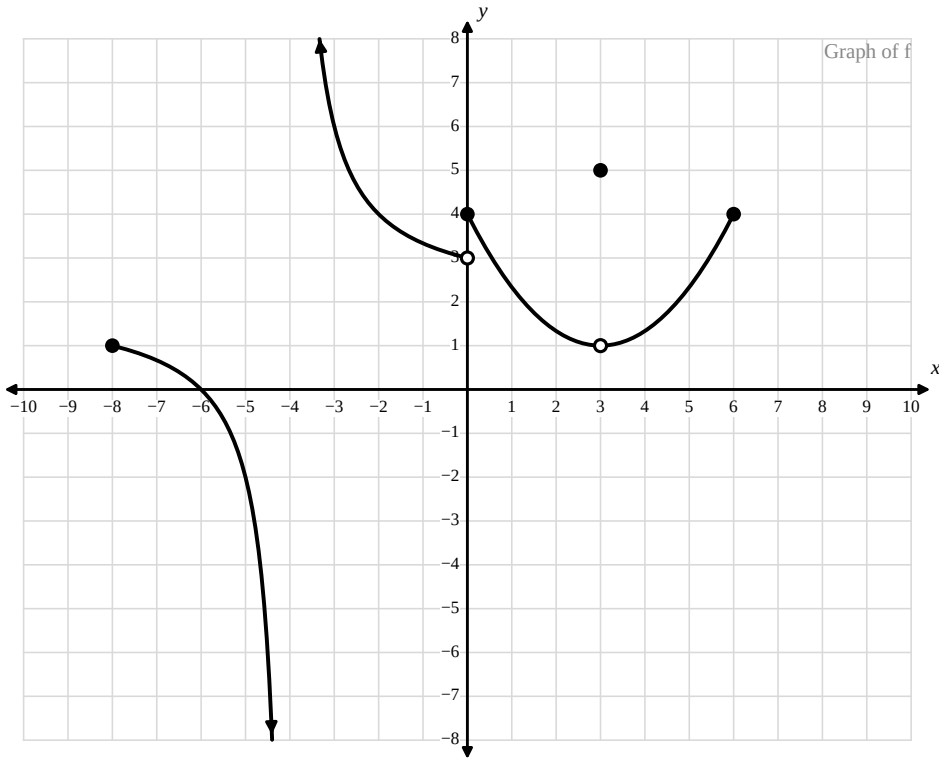
PRACTICE D · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

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SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)



1. $\lim_{x\to 3} f(x) =$

Both sides head to the open circle at $(3, 1)$.

1

2. $\lim_{x\to -4^-} f(x) =$

Just left of $x = -4$ *the curve falls without bound.*

$-\infty$

3. $\lim_{x\to 0} f(x) =$

$\lim_{x\to 0^-} f(x) = 3$, $\lim_{x\to 0^+} f(x) = 4$

$3 \neq 4$, *so* $\lim_{x\to 0} f(x)$ *does not exist.*

DNE

(2 points)

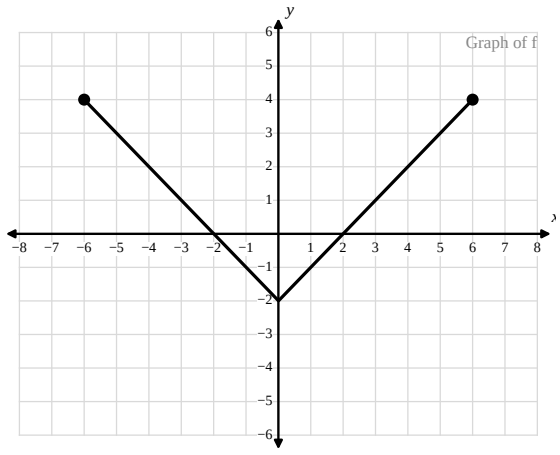
4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = 4$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

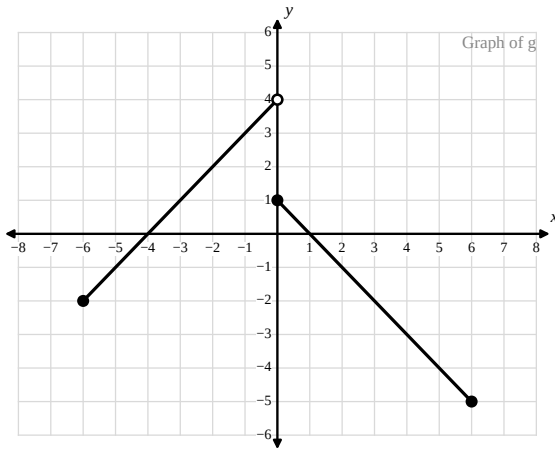
Height 4 is reached from both sides only at $x = -2$.

Not $x = 0$ *(right side only; the left heads to 3) and not* $x = 6$ *(the graph ends there).*

$a = -2$

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim\limits_{x\to 0^-} \dfrac{[f(x)]^2 + g(x)}{f(x) - g(x)}$

$$\lim_{x\to 0^-} \frac{[f(x)]^2 + g(x)}{f(x) - g(x)} = \frac{[\lim_{x\to 0^-} f(x)]^2 + \lim_{x\to 0^-} g(x)}{\lim_{x\to 0^-} f(x) - \lim_{x\to 0^-} g(x)}$$

Quotient (bottom limit is −6, not 0), power, sum, difference.

$$= \frac{(-2)^2 + 4}{-2 - 4} = \frac{8}{-6} = -\frac{4}{3}$$

$-\frac{4}{3}$

(2 points)

6. Use the table to estimate $\lim\limits_{x\to 0} k(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	−0.1	−0.01	−0.001	→ 0 ←	0.001	0.01	0.1
$k(x)$	0.951626	0.995017	0.999500		1.000500	1.005017	1.051709

Left: 0.951626, 0.995017, 0.999500

Right: 1.051709, 1.005017, 1.000500

Both sides agree. A table estimates; it does not prove.

≈ 1

$\lim\limits_{x\to 0^-} k(x) \approx$ 1 $\lim\limits_{x\to 0^+} k(x) \approx$ 1 $\lim\limits_{x\to 0} k(x) \approx$ 1

Evaluate each limit. (2 points each)

7. $\lim_{x\to\frac{\pi}{6}} \left(\sec^2(2x) - 4\sin x\right)$

$\lim_{x\to\frac{\pi}{6}} \left(\sec^2(2x) - 4\sin x\right) = \sec^2\frac{\pi}{3} - 4\sin\frac{\pi}{6}$

$= (2)^2 - 4\left(\frac{1}{2}\right) = 4 - 2 = 2$

2

8. $\lim_{x\to -2} \dfrac{x^3 + 2x^2 - x - 2}{x^2 + 3x + 2}$

Substitution produces 0/0: indeterminate.

Factor the top by grouping.

$\lim_{x\to -2} \dfrac{x^2(x+2) - (x+2)}{(x+1)(x+2)}$

$= \lim_{x\to -2} \dfrac{(x+2)\left(x^2-1\right)}{(x+1)(x+2)}$

$= \lim_{x\to -2} \dfrac{(x-1)(x+1)}{x+1} = \lim_{x\to -2}(x-1) = -3$

-3

9. $\lim_{h\to 0} \dfrac{\sqrt{9+h} - 3}{h^2 + h}$

Substitution produces 0/0: indeterminate.

$\lim_{h\to 0} \dfrac{(9+h) - 9}{h(h+1)\left(\sqrt{9+h}+3\right)}$

$= \lim_{h\to 0} \dfrac{1}{(h+1)\left(\sqrt{9+h}+3\right)}$

$= \dfrac{1}{(1)(3+3)} = \dfrac{1}{6}$

$\frac{1}{6}$

10. $\lim_{x\to 0} \dfrac{\frac{1}{(x+2)^2} - \frac{1}{4}}{x}$

Substitution produces 0/0: indeterminate.

LCD: 4(x + 2)²

$\lim_{x\to 0} \dfrac{4 - (x+2)^2}{4x(x+2)^2}$

$= \lim_{x\to 0} \dfrac{-x^2 - 4x}{4x(x+2)^2} = \lim_{x\to 0} \dfrac{-x(x+4)}{4x(x+2)^2}$

$= \lim_{x\to 0} \dfrac{-(x+4)}{4(x+2)^2} = \dfrac{-4}{16} = -\dfrac{1}{4}$

$-\frac{1}{4}$

11. For $f(x) = \dfrac{1}{x^2}$, find $\lim_{h\to 0} \dfrac{f(x+h) - f(x)}{h}$

$\lim_{h\to 0} \dfrac{\frac{1}{(x+h)^2} - \frac{1}{x^2}}{h}$

LCD: x²(x + h)²

$= \lim_{h\to 0} \dfrac{x^2 - (x+h)^2}{h\,x^2(x+h)^2}$

$= \lim_{h\to 0} \dfrac{-2xh - h^2}{h\,x^2(x+h)^2} = \lim_{h\to 0} \dfrac{-2x - h}{x^2(x+h)^2}$

$= \dfrac{-2x}{x^4} = -\dfrac{2}{x^3}$

$-\frac{2}{x^3}$